

Due: Thursday, September 7th at 9:50 am.

From *Levin*, please reread Section 0.2 for the quantifier information. Then read Section 0.3 and complete the following

- Chapter 0, Section 2: #12 - 15
 - Chapter 0, Section 3: #1 - 10
 - Complete the Trials of Xanthius.
 - And complete the following:
- (1) Let $A = \mathbb{N}$, $B = \{n \in \mathbb{N} \mid n \leq 10\}$, and C be the positive even numbers.
- (a) In words, describe the set B and, if possible, write the whole set out.
 - (b) Write set C using set builder notation.
 - (c) Is $B \subseteq A$?
 - (d) Is $B \subset A$?
 - (e) Is $C \subseteq B$?
 - (f) Is $\{1, 2, 3, 4\} \subseteq A$?
 - (g) Is $\{1, 2, 3, 4\} \subseteq B$?
 - (h) Is $\{1, 2, 3, 4\} \subseteq C$?
 - (i) What elements are in both A and B ?
 - (j) What elements are in both B and C ?
- (2) For the following descriptions of sets, write them using set builder notation.
- (a) The perfect squares: 1, 4, 9, 16, 25, ...
 - (b) The numbers that are rational but are not integers.
 - (c) The ordered pairs where the y is larger than the x value.
 - (d) The ordered pairs that are all the points on the parabola $y = x^2$.